



Teradata Factory Xpress

Review Questions & Lab Exercises Workbook

Course # 56749

Version 16.20.4



Student Lab Guide

Teradata Learning

Notes

Table of Contents

Module Exercises & Review Questions.....	4
Lab Environment.....	58
Lab Exercise 7-1	68
Lab Exercise 11-1	76
Lab Exercise 11-2	80
Lab Exercise 11-3 (optional).....	82
Lab Exercise 14-1	86
Lab Exercise 15-1	94
Lab Exercise 18-1	102
Lab Exercise 18-2	106
Lab Exercise 20-1	110
Lab Exercise 20-2	112
Lab Exercise 23-1	120
Lab Exercise 23-2 (optional).....	126

Module Exercises & Review Questions

This workbook section contains module exercises and the review questions for the Teradata Factory Xpress class. This paper copy is included to facilitate answering questions on paper instead of in the electronic format.

This section contains module exercises and review questions for the Teradata Factory Xpress class.

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Module 1: Review Questions

1. Which feature allows the Teradata Database to process enormous volumes of data quickly? ____
 - a. High availability software and hardware components
 - b. High performance servers from Intel
 - c. Proven Scalability
 - d. Parallelism
2. The Teradata Database is primarily a ____ .
 - a. Client
 - b. Server
3. Which choice represents a quadrillion bytes or a Petabyte (PB) of data? ____
 - a. 10^9
 - b. 10^{12}
 - c. 10^{15}
 - d. 10^{18}
4. In a relational table, the set of columns that uniquely identifies a row is the ____ .

Module 2: Review Questions

1. What language is used to access a Teradata table?
2. What are five Teradata database objects?
3. What are four major components of the Teradata architecture?
4. What are views?
5. What are macros?

Module 3: Review Questions

1. What are the two software elements that allow a client application to connect with a PE?
2. What is the purpose of the PE?
3. What is the purpose of the AMP?
4. How many sessions can a PE support?

Match Quiz

- | | |
|------------------------------|---|
| ___ 1. CLI | a. Does Aggregating and Locking |
| ___ 2. MTDP | b. Validates SQL syntax |
| ___ 3. MOSI | c. Connects AMPs and PEs |
| ___ 4. Parser | d. Balances sessions across PEs |
| ___ 5. AMP | e. Provides Client side OS independence |
| ___ 6. Message Passing Layer | f. Library of Session Management Routines |
| ___ 7. TDP | g. PE S/W that chooses the execution plan |
| ___ 8. Optimizer | h. PE S/W sends plan steps to AMP(s) |
| ___ 9. Dispatcher | i. Library of Teradata Service Routines |
| ___ 10. Parallelism | j. Foundation of Teradata architecture |

Module 4: Review Questions

True or False

- _____ 1. A database will always have tables.
 - _____ 2. A user will always have a password.
 - _____ 3. A user creating a subordinate user must give up some of his/her Perm Space.
 - _____ 4. Creating tables requires the definition of at least 1 column and a Primary Index.
 - _____ 5. The sum of all user and database max Perm Space will equal the total space on the system.
 - _____ 6. The sum of all user and database Spool Space will equal the total space on the system.
 - _____ 7. The limit of space for both Volatile and Global Temporary tables is based on the maximum TEMPORARY space for the user.
 - _____ 8. Deleting a macro from a database reclaims Perm Space for the database.
9. Which two statements are TRUE about PERM space in Teradata 16.20? (choose two) _____
- a. PERM space cannot be dynamically modified.
 - b. The per/AMP limit of PERM space can be exceeded.
 - c. Tables, index subtables, and stored procedures use PERM space.
 - d. Maximum PERM space can be defined at the database or table level.
10. Which statement is TRUE about SPOOL space in Teradata 16.20? _____
- a. SPOOL space cannot be dynamically modified.
 - b. Maximum SPOOL space can be defined at the database or user level.
 - c. The SPOOL limit is dependent on the database limit where the table is located.
 - d. Maximum SPOOL space can be defined at a value greater than the creator's value.

Module 5: Review Questions

Answer the following either as True or False as these apply to Primary Indexes:

- True or False 1. UPI and NUPI equality value accesses are always a one-AMP operation.
- True or False 2. UPI and NUPI indexes allow NULL in a primary index column.
- True or False 3. UPI, NUPI, and NOPI tables allow duplicate rows in the table.
- True or False 4. A UPI can be used as a Primary Key implementation.

Fill in the Blanks

5. The output of the hashing algorithm is called the _____.
6. To determine the target AMP, the Message Passing Layer must lookup an entry in the Hash Map based on the _____.
7. A Row ID consists of a row hash plus a _____ value.
8. A uniqueness value is required to produce a unique Row ID because of _____ and _____.
9. Once the target AMP has been determined for a PI search, the _____ for that AMP is accessed to determine the cylinder that may hold the row.
10. The Cylinder Index points us to the address and length of the data _____.

Module 6: Review Questions

Fill each box with either Yes, No, or the appropriate number.

	USI Access	NUSI Access	FTS
# AMPs			
# rows			
Parallel Operation			
Uses Hash Maps			
Uses Separate Sub-table			
Reads all data blocks of table			

Module 7: Review Questions

1. Which data interface is available with Teradata Studio?
 - a. CLIV2
 - b. JDBC
 - c. ODBC
 - d. Teradata .Net
2. Which piece of query information is not available in the History Window for Teradata Studio?
 - a. User name
 - b. Query band
 - c. Elapsed time
 - d. Data source name
 - e. Number of rows returned
3. What is a technique to execute multiple statements as a multi-statement request in Teradata Studio?

Module 8: Review Questions

Complete the following.

1. Each AMP has its own memory and manages its own disk space and executes independently of other AMPs. This is referred to as a _____ architecture.
2. The software component that allows the Teradata Database to execute in different operating system environments is the _____.
3. A physical message passing interconnect is called the _____.
4. A clique provides protection from a _____ failure.
5. If a node fails, all vprocs will migrate to the remaining nodes in the clique. This feature is referred to as _____.
6. The _____ provides a single point of operational control for Teradata MPP systems.
7. A _____ node is part of a system configuration, is connected to the BYNET, and executes the Teradata Database software.
8. A _____ node is part of a system configuration, may connect to the BYNET, and is used to execute application software other than Teradata Database software.
9. A _____ node is part of a system configuration, connects to the BYNET, and is used as a spare node in the event of a node failure.

Module 9: Review Questions

1. What is the WAL Depot used for?
 - a. UNDO Rows
 - b. New data blocks
 - c. Master Index updates
 - d. Write-in-place data blocks
2. Which two are placed into the WAL Log?
 - a. REDO Rows
 - b. UNDO Rows
 - c. New data blocks
 - d. Write-in-place data blocks

Match the item to a lettered description.

- | | |
|----------------------------|--|
| _____ 1. Database locks | a. Provides for TXN rollback in case of failure |
| _____ 2. Table locks | b. Teradata Backup and Recovery applications |
| _____ 3. Row Hash locks | c. Protects all rows of a table |
| _____ 4. FALLBACK | d. Logs changed rows for down AMP |
| _____ 5. Cluster | e. Provides for recovery to a point in time |
| _____ 6. Recovery journal | f. Applies to all tables and views within |
| _____ 7. Transient journal | g. Lowest level of protection granularity |
| _____ 8. NetBackup/Tivoli | h. Protects tables from AMP failure |
| _____ 9. Permanent journal | i. Protects database from a physical drive failure |
| _____ 10. Disk Array | j. Group of AMPs used by Fallback |

Module 10: Review Questions

1. Which two are accessed in FSG cache memory? (choose 2)
 - a. Hash maps
 - b. Master Indexes
 - c. Cylinder Indexes
 - d. Data Dictionary Cache
2. What are two benefits of Cylinder Index Segregation in Teradata 15.10? (choose 2)
 - a. Allows for larger cylinders
 - b. Improves Teradata reboot performance
 - c. Allows more user data to be stored in a cylinder
 - d. Reduces the number of required cylinder indexes
3. When are multiple partitions in a LUN used in a Teradata configuration? (choose 2)
 - a. SSD drives with a 2850
 - b. HDD drives with a 2850
 - c. SSD drives with a 6800
 - d. HDD drives with a 6800
4. Describe the fundamental relationship between Linux, logical units, and disk array controllers.

5. Describe the fundamental relationship between AMPs, Vdisks, Pdisks, Partitions, and LUNs.

Module 11: Order of Preference Exercise

Assuming the system default is "Primary Index", show the indexes that are created as a result of the DDL.

```
CREATE TABLE table_1
(col1 INTEGER NOT NULL UNIQUE          col1 =
,col2 INTEGER NOT NULL PRIMARY KEY);  col2 =
```

UPI =	Unique Primary Index
NUPI =	Non Unique Primary Index
USI =	Unique Secondary Index

```
CREATE TABLE table_2
(col1 INTEGER NOT NULL PRIMARY KEY      col1 =
,col2 INTEGER)                          col2 =
PRIMARY INDEX (col2);
```

```
CREATE TABLE table_3
(col1 INTEGER                          col1 =
,col2 INTEGER NOT NULL);               col2 =
```

```
CREATE TABLE table_4
(col1 INTEGER NOT NULL                col1 =
,col2 INTEGER NOT NULL                col2 =
,col3 INTEGER NOT NULL UNIQUE         col3 =
,CONSTRAINT pk1 PRIMARY KEY (col1,col2)); (col1,col2) =
```

```
CREATE TABLE table_5
(col1 INTEGER NOT NULL                col1 =
,col2 INTEGER NOT NULL                col2 =
,col3 INTEGER NOT NULL UNIQUE         col3 =
,CONSTRAINT uniq1 UNIQUE (col1,col2)); (col1,col2) =
```

Module 11: Exercise 2 – Example

On the following pages, there are sample tables with distribution demographics.

- Indicate ALL possible Primary Index candidates (UPI and NUPI).
- Later exercises will guide your final choices.

Primary Index Candidate Guidelines:

- PK and UNIQUE COLUMNS (ND)
- Any single column with:
 - High Distinct values (at least 10X)
 - Low Maximums for NULLs or a Value
 - Typical Rows that is close to Max Rows

300,000,000 Rows	A	B	C	D	E	F	G	H	
PK/FK	PK,SA		FK,NN	NN,ND					
Distinct Values	300M	35M	7.5M	300M	8	15M	15M	700	
Max Rows/Value	1	12	500	1	40M	50	4M	450K	
Max Rows/NULL	0	5	0	0	0	4M	0	50K	
Typical Rows/Value	1	7	35	1	35M	15	15	400K	
PI/SI	UPI	NUPI	NUPI?	UPI					
Collect Statistics (Y/N)									

Exercise 2 – Choosing PI Candidates

ENTITY 1		A	B	C	D	E	F
500,000,000 Rows							
PK/FK	PK,UA						
Distinct Values	500M	475M	1.5M	1.25M	200M	5M	
Max Rows/Value	1	2	400	450	3	110	
Max Rows/NULL	0	0	0	0	10M	0	
Typical Rows/Value	1	1	325	350	2	90	
PI/SI							
Collect Statistics (Y/N)							

Exercise 2 – Choosing PI Candidates (cont.)

		DEPENDENT					
5,000,000 Rows		A	M	N	O	P	Q
PK/FK		PK				NN, ND	
		FK	SA				
Distinct Values		2M	50	90K	3M	5M	2M
Max Rows/Value		4	200K	75	2	1	5
Max Rows/NULL		0	0	0	390K	0	1M
Typical Rows/Value		1	60K	50	1	1	1
	PI/SI						
Collect Statistics (Y/N)							

Exercise 2 – Choosing PI Candidates (cont.)

		ASSOCIATIVE 1			
1.5 Billion Rows		A	G	R	S
PK/FK		PK			
		FK	FK,SA		
	Distinct Values	490M	50M	75K	4M
	Max Rows/Value	5	40	21K	400
	Max Rows/NULL	0	0	0	0
	Typical Rows/Value	3	25	19K	350
	PI/SI				
Collect Statistics (Y/N)					

Exercise 2 – Choosing PI Candidates (cont.)

ASSOCIATIVE 2							
500,000,000 Rows	<table border="1"> <thead> <tr> <th>A</th> <th>M</th> <th>G</th> <th>T</th> <th>U</th> <th></th> </tr> </thead> </table>	A	M	G	T	U	
A	M	G	T	U			
PK/FK	PK						
	FK						
Distinct Values	250M						
Max Rows/Value	3						
Max Rows/NULL	0						
Typical Rows/Value	1						
PI/SI							
Collect Statistics (Y/N)							

Exercise 2 – Choosing PI Candidates (cont.)

		HISTORY					
3.65 Billion Rows		A	DATE	D	E	F	
PK/FK		PK					
	FK	SA					
Distinct Values	500M	730	N/A	N/A	N/A		
Max Rows/Value	18	5.5M	N/A	N/A	N/A		
Max Rows/NULL	0	0	N/A	N/A	N/A		
Typical Rows/Value	3	4.5M	N/A	N/A	N/A		
	PI/SI						
Collect Statistics (Y/N)							

Module 11: Review Questions

1. Which trade-off must be balanced to make the best choice for a primary index? ____
 - a. Access and volatility
 - b. Access and block size
 - c. Block size and volatility
 - d. Access and distribution

2. When volatility is considered as one of the Primary Index choice criteria, what is analyzed? ____
 - a. Degree of uniqueness
 - b. How often the data values will change
 - c. How often the fixed length rows will change
 - d. How frequently the column is used for access

3. To optimize the use of disk space, the designer should choose a primary index that _____.
 - a. is non-unique
 - b. consists of one column
 - c. is unique or nearly unique
 - d. consists of multiple columns
 - e. has fewer distinct values than AMPs

Module 11: Review Questions (cont.)

4. For NoPI tables, what are 2 ways in which the Random Generator is executed?
 - a. At the PE level for ad hoc SQL requests
 - b. At the TPT Stream client level for array insert operations
 - c. At the PE level with TPT Stream for array insert operations
 - d. At the AMP level for INSERT-SELECT into an empty NoPI table

5. Assume DBSControl flag #53 (Primary Index Default) is set to N (No Primary Index), which two indexes are created for TableX given the following DDL command?


```
CREATE TABLE TableX
      (col1    INTEGER NOT NULL UNIQUE
      ,col2    CHAR(10) NOT NULL PRIMARY KEY
      ,col3    CHAR(80));
```

 - a. col1 will be a UPI
 - b. col1 will be a USI
 - c. col2 will be a UPI
 - d. col2 will be a USI

6. Which two options are permitted for NoPI tables?
 - a. Fallback
 - b. TPT Update
 - c. Hash Index
 - d. BLOBs and CLOBs

Module 12: Review Questions

1. The Row Hash for a PI value of 824 is the same for the data types of INTEGER and DECIMAL(18,0). True or False. _____
2. The first 20 bits of the Row Hash is referred to as the _____.
3. The Hash Map consists of entries or buckets which identify an _____ number for the Row Hash.
4. The Current Configuration _____ Hash Map is used to locate the AMP to locate/store a row based on PI value.
5. The _____ utility is used to redistribute rows to a new system configuration with more AMPs.
6. When creating a new table, the Unique Value of a Table ID comes from the dictionary table named DBC._____.
7. The Row ID consists of the _____ and the _____.
8. The _____ contains a Cylinder Index Descriptor (CID) for each allocated Cylinder.
9. The _____ contains an entry for each data block in the cylinder.
10. The _____ consists of a set of pointers to the data rows in data block.
11. In TD 14.0, the maximum block size is approximately _____ and the maximum row size is approximately _____.
12. In TD 16.00, the maximum block size is approximately _____ and the maximum row size is approximately _____.

Module 13: Review Questions

1. When Teradata INSERTs a new row into a table, it first goes to the _____ to locate the proper cylinder for the new row.
 - a. Cylinder Index
 - b. Fallback AMP
 - c. Free Cylinder List
 - d. Master Index
2. When a new block is needed, the File System searches the Free Block List looking for the first Free Block whose size is equal to, or greater than the new block's requirement. It does not have to be an exact match.
 - a. True
 - b. False
3. Name the condition which occurs when there is no block on the Free Block List with enough sectors to accommodate the additional data during an INSERT or UPDATE.
 - a. Mini Cylinder Pack
 - b. Cylinder Migrate to a new cylinder
 - c. Cylinder Migrate to an adjacent cylinder
 - d. Cylinder Full
4. The _____ parameter can be set to control how completely cylinders are filled during loading and PackDisk.
 - a. FreeSpace
 - b. PermDBSize
 - c. DataBlockSize
 - d. PermDBAllocUnit

Module 13: Review Questions (cont.)

5. Number the following steps in sequence from 1 to 6 that the File System software will attempt to perform in order to insert a new row into an existing data block.
- _____ Perform a Cylinder Migrate operation to an adjacent cylinder
 - _____ Simply insert the row into data block if enough contiguous free bytes in the block
 - _____ Perform a Block split
 - _____ Perform a Cylinder Migrate operation to a new cylinder
 - _____ Defragment the block and insert the row
 - _____ Expand or grow the block to hold the row
6. As part of a cylinder full condition, if the number of free sectors within a cylinder is greater than 25%, what operation will Teradata perform in the background? _____
7. If the number of free cylinders falls below a minimum threshold, what operation will Teradata perform in the background? _____

Module 14: Review Questions

1. The Row ID for a partitioned table consists of _____ + _____ + _____.
2. The Row Key consists of the _____ + _____.
3. With a non-partitioned table, the partition number defaults to _____.
4. True or False. For a partitioned table, the partition number and the Row Hash are both used by the Message Passing Layer to determine which AMP(s) should receive the request.
5. Which two options apply to the RANGE_N expression in a partitioning expression? _____
 - a. Ranges can be listed in descending order
 - b. Allows use of NO RANGE OR UNKNOWN option
 - c. Partitioning column must be part of the Primary Index
 - d. Maximum of 9.223 quintillion partitions with Teradata 16.20
6. With a populated table, which two actions are allowed with the ALTER TABLE command. _____
 - a. Drop all of the ranges
 - b. Add or drop ranges from the partition "ends"
 - c. Change the columns that comprise the primary index
 - d. Add or drop special partitions (NO RANGE, UNKNOWN)
7. What are two advantages of partitioning a table? _____
 - a. Fast delete of rows in partitions
 - b. Fewer AMPs are involved when accessing data
 - c. Range queries can be executed without doing an all-rows scan
 - d. Faster access for a PI value access than for a non-partitioned table

Module 14: Review Questions (cont.)

Given this CREATE TABLE statement, answer the following questions.

```
CREATE TABLE Orders
  (Orderid          INTEGER NOT NULL,
   Custid           INTEGER NOT NULL,
   Orderstatus      CHAR(1),
   Ordertotal       DECIMAL(9,2) NOT NULL,
   Orderdate        DATE NOT NULL FORMAT 'YYYY-MM-DD',
   Orderpriority    SMALLINT,
   Clerk            CHAR(16),
   Shippriority     SMALLINT,
   Ordercomment     VARCHAR(80))
PRIMARY INDEX (Orderid)
PARTITION BY RANGE_N (Orderdate BETWEEN
  DATE '2014-01-01' AND DATE '2020-12-31' EACH INTERVAL '1' DAY)
UNIQUE INDEX (Orderid);
```

8. What is the name of partitioning column? _____
9. What is the time period for each partition? _____
10. What are two reasons that a USI is specified instead of defining Orderid as a UPI? _____
 - a. This is a coding style choice.
 - b. You cannot have a UPI when using a partitioned primary index.
 - c. To improve the performance of value access queries for a specific Orderid.
 - d. You cannot have a UPI if the partitioning column is not part of the primary index.

Module 15: Review Questions

1. Which choice is NOT allowed with a column partitioned table in Teradata 16.20?
 - a. SET table
 - b. NoPI table
 - c. MULTISSET table
 - d. Table with a NUPI
2. What are two benefits of Column Partitioning? (choose two)
 - a. Reduced I/O
 - b. Reduced CPU
 - c. Reduced disk space usage
 - d. Reduced tactical query response times
3. True or False? Deleting a row in a column partitioned table will reclaim table space.
4. True or False? In a multilevel columnar table, only one level may be defined with COLUMN.
5. True or False? The preferred way to load a columnar table is using INSERT/SELECT.

Module 16: Review Questions

1. Because the row is hash-distributed on different columns, the subtable row will typically land on an AMP other than the one containing the data row. This index would be:
 - a. UPI or NUPI
 - b. USI
 - c. NUSI
 - d. None of the above
2. The Teradata DBS hashes the value and uses the Row Hash to find the desired rows. This is always a one-AMP operation. This index would be:
 - a. UPI or NUPI
 - b. USI
 - c. NUSI
 - d. None of the above
3. _____ is a process that determines common Row IDs between multiple NUSI values by a process of intersection.
 - a. NUSI Bit Mapping
 - b. Dual NUSI Access
 - c. Full Table Scan
 - d. NUSI Read
4. If aggregation is performed on a NUSI column, the Optimizer accesses the NUSI subtable and returns the result without accessing the base table. This is referred to as:
 - a. NUSI bit mapping
 - b. Full table scan
 - c. Dual NUSI access
 - d. Covering Index

Module 17: Exercise 3 – Example

The sample tables now include value/range and change row demographics.

- Delete any PI candidates that are too volatile.
- Identify any USI candidates.
- Identify any NUSI or VONUSI candidates if typical rows/value is $\leq 5\%$ or # of distinct values ≥ 20 AND Value/Range Access > 0 .
- Later exercises will guide your final choices.

Change Rating Guidelines:

- PI – change rating 0 - 1.
- SI – change rating 0 - 3.

Value Access Guideline:

- NUSI Value Access > 0
- VONUSI Range Access > 0

300,000,000 Rows	A	B	C	D	E	F	G	H	
PK/FK	PK,SA		FK,NN	NN,ND					
Value Access	5K	26K	0	500K	0	0	0	260	
Range Access	12	0	0	0	0	0	0	4K	
Distinct Values	300M	35M	7.5M	300M	8	15M	15M	700	
Max Rows/Value	1	12	500	1	40M	50	4M	450K	
Max Rows/NULL	0	5	0	0	0	4M	0	50K	
Typical Rows/Value	1	7	35	1	35M	15	15	400K	
Change Rating	0	1	4	2	0	4	4	9	
PI/SI	UPI	NUPI	NUPI	UPI					
	USI	NUSI		USI					
Collect Statistics (Y/N)									

Exercise 3 – Choosing SI Candidates (cont.)

ENTITY 1							
500,000,000 Rows	A	B	C	D	E	F	
PK/FK	PK,UA						
Value Access	250K	0	0	0	0	0	
Range Access	0	0	0	0	0	0	
Distinct Values	500M	475M	1.5M	1.25M	200M	5M	
Max Rows/Value	1	2	400	450	3	110	
Max Rows/NULL	0	0	0	0	10M	0	
Typical Rows/Value	1	1	325	350	2	90	
Change Rating	0	2	1	1	1	1	
PI/SI	UPI	NUPI	NUPI	NUPI		NUPI	
Collect Statistics (Y/N)							

Exercise 3 – Choosing SI Candidates (cont.)

		ENTITY 2					
50,000,000 Rows		G	H	I	J	K	L
PK/FK		PK,SA					
Value Access		5K	365	52	12	0	0
Range Access		12	0	0	0	0	2.6K
Distinct Values		50M	500K	45M	12	50	900K
Max Rows/Value		1	200	2	5M	1.2M	60
Max Rows/NULL		0	0	450K	0	0	0
Typical Rows/Value		1	100	1	4M	950K	50
Change Rating		0	0	9	1	2	0
PI/SI		UPI	NUPI				NUPI
Collect Statistics (Y/N)							

Exercise 3 – Choosing SI Candidates (cont.)

5,000,000 Rows	DEPENDENT						
	A	M	N	O	P	Q	
PK/FK	PK				NN,ND		
	FK	SA					
Value Access	0	0	0	0	0	0	
Range Access	0	0	0	0	0	0	
Distinct Values	2M	50	90K	3M	5M	2M	
Max Rows/Value	4	200K	75	2	1	5	
Max Rows/NULL	0	0	0	390K	0	1M	
Typical Rows/Value	1	60K	50	1	1	1	
Change Rating	0	0	3	1	0	1	
PI/SI	UPI				UPI		
	NUPI		NUPI				
Collect Statistics (Y/N)							

Exercise 3 – Choosing SI Candidates (cont.)

ASSOCIATIVE 1					
1.5 Billion Rows	A	G	R	S	
PK/FK	PK				
	FK	FK,SA			
Value Access	26K	0	0	0	
Range Access	0	0	0	0	
Distinct Values	490M	50M	75K	4M	
Max Rows/Value	5	40	21K	400	
Max Rows/NULL	0	0	0	0	
Typical Rows/Value	3	25	19K	350	
Change Rating	0	0	0	0	
PI/SI	UPI				
	NUPI	NUPI	NUPI?	NUPI	
Collect Statistics (Y/N)					

Exercise 3 – Choosing SI Candidates (cont.)

ASSOCIATIVE 2						
500,000,000 Rows	A	M	G	T	U	
PK/FK	PK					
	FK		FK			
Value Access	0		0	0	0	
Range Access	0		0	0	0	
Distinct Values	250M		50M	2.8M	750	
Max Rows/Value	3		150	180	675K	
Max Rows/NULL	0		0	0	0	
Typical Rows/Value	1		8	170	500K	
Change Rating	0		0	0	0	
PI/SI	UPI					
	NUPI		NUPI	NUPI		
Collect Statistics (Y/N)						

Exercise 3 – Choosing SI Candidates (cont.)

HISTORY						
3.65 Billion Rows	A	DATE	D	E	F	
PK/FK	PK					
	FK	SA				
Value Access	10M	26K	0	0	0	
Range Access	0	100K	0	0	0	
Distinct Values	500M	730	N/A	N/A	N/A	
Max Rows/Value	18	5.5M	N/A	N/A	N/A	
Max Rows/NULL	0	0	N/A	N/A	N/A	
Typical Rows/Value	3	4.5M	N/A	N/A	N/A	
Change Rating	0	0	N/A	N/A	N/A	
PI/SI	UPI					
	NUPI					
Collect Statistics (Y/N)						

Module 17: Review Questions

1. With a NUPI, a technique to avoid a duplicate row check is to _____.
 - a. use set tables
 - b. use NOT NULL on the NUPI column(s)
 - c. create the table as a MULTiset table
 - d. compare data values byte-by-byte within a Row Hash to ensure uniqueness
2. Which type of usage normally applies to a USI? ____
 - a. Range access
 - b. NOT condition
 - c. Equality value access
 - d. Inequality value access
3. Which two types of usage normally apply to a composite NUSI that is hash-ordered? ____ ____
 - a. Covering index
 - b. Equality value access
 - c. Inequality value access
 - d. Non-covering range access
4. Which situation requires placing before-images of rows into the Transient Journal? ____
 - a. loading a table with TPT Load
 - b. DELETE all the rows in a table
 - c. UPDATE all the rows in a table
 - d. INSERT/SELECT into an empty table

Module 18: Review Questions

1. Which choice can be used with the COMPRESS option?
 - a. Identity column
 - b. Non-unique Primary Index
 - c. USI as PK with Standard RI
 - d. Non-unique Secondary Index
2. Which section of a row identifies the starting location of variable length column data and is present only if variable length columns without compression are declared?
 - a. Presence Bits
 - b. Column Offsets
 - c. VARCHAR Columns
 - d. Uncompressed Columns
3. How can you override the default that a column with a NULL value will require row space?
 - a. Use the NOT NULL option on the column as part of the CREATE TABLE statement.
 - b. Set the user's default so columns will default to COMPRESS when creating a table.
 - c. Use the COMPRESS option on the column as part of the CREATE TABLE statement.
 - d. Use the DEFAULT NULL option on the column as part of the CREATE TABLE statement.
4. What is the minimum space the table headers will take for a 6-column table on a 10 AMP system that has block level compression?
 - a. 4,096 bytes
 - b. 8,192 bytes
 - c. 32,768 bytes
 - d. 40,960 bytes
5. What DD view is used to get sizing information about tables? _____

Module 19: Review Questions

1. Where are statistics stored in Teradata 16.20 when you COLLECT STATISTICS on a NUPI? _____
 - a. DBC.StatsTbl
 - b. DBC.Indexes
 - c. DBC.TVFields
 - d. Data Dictionary cache
2. Random AMP sample statistics are stored in _____.
 - a. DBC.Indexes
 - b. DBC.TVFields
 - c. Data Dictionary cache
 - d. Request-to-Steps cache
3. Which statement is used to display statistical detail information (e.g., summary section, high bias values, and intervals)? _____
 - a. HELP INDEX
 - b. SHOW TABLE
 - c. HELP STATISTICS
 - d. SHOW STATISTICS
4. Which statement is used to display extrapolated statistics used by the optimizer? _____
 - a. HELP INDEX
 - b. HELP STATISTICS
 - c. HELP CURRENT STATISTICS
 - d. SHOW CURRENT STATISTICS

Module 20: Review Questions

Fill in the blanks.

1. _____ steps are multi-AMP processing steps that are numbered but execute at the same time.
2. The primary way to help the Optimizer make the best choices and ensure the most accurate EXPLAIN output is to _____ on appropriate indexes and columns.
3. An EXPLAIN plan will indicate “estimated with _____ confidence” when a value for an index is provided to retrieve the data and the index has collected statistics.

Module 21: Review Questions

1. The best way to be sure what type of Join will occur is to use the EXPLAIN facility.
 - a. True
 - b. False

2. When two tables are to be Merge Joined, which is the best case of the following scenarios :
 - a. The Join column(s) are the Primary Index of both tables.
 - b. The Join column(s) are not a Primary Index of either table.
 - c. The Join column(s) are the Primary Index of one of the tables.
 - d. The Join column(s) are a Unique Secondary Index on both of the tables.

3. Match the four join plans with its correct description.

___ Product Join	a. Most efficient type of Join; the only type of join that doesn't always use all of the AMPs. The number of AMPs involved will vary.
___ Merge Join	b. Based on set subtraction; used for finding rows that don't have a matching row in the other table. Queries with the NOT IN and EXCEPT operator lead to this type of join.
___ Nested Join	c. Every qualifying row of one table is compared to every qualifying row in the other table. Rows that match on their WHERE conditions are then saved.
___ Exclusion Join	d. Commonly done when the join condition is based on equality. Efficient because every row is not compared with every row in other table.

Module 21: Review Questions (cont.)

Fill in the blanks.

4. When joining two row partitioned tables that are not partitioned in the same manner, a technique available to the optimizer is referred to as the _____ window.
5. A direct merge join of two row partitioned tables when the tables have the same PI and identical partitioning expressions is referred to as a _____ - based merge join.
6. The term _____ _____ refers to an automatic optimization in which the optimizer determines, based on query conditions, that some partitions can be skipped.

Exercise 4 – Example

On the following pages, there are sample tables with change row and value access demographics.

- Make your final index choices, as shown below.
- Identify columns to collect statistics on.
- Join Access demographics have been added.

Final Index Choice Guidelines:

- Choose ONE Primary Index (UPI or NUPI) utilizing Join Access demographics.
- Do not carry identical Primary and Secondary Indexes.

300,000,000 Rows	A	B	C	D	E	F	G	H
PK/FK	PK,SA		FK,NN	NN,ND				
Value Access	5K	26K	0	500K	0	0	0	260
Range Access	12	0	0	0	0	0	0	4K
Join Access	1M	0	2.6K	0	0	0	0	0
Join Rows	50M	0	65K	0	0	0	0	0
Distinct Values	300M	35M	7.5M	300M	8	15M	15M	700
Max Rows/Value	1	12	500	1	40M	50	4M	450K
Max Rows/NULL	0	5	0	0	0	4M	0	50K
Typical Rows/Value	1	7	35	1	35M	15	15	400K
Change Rating	0	1	4	2	0	4	4	9
PI/SI	UPI	NUPI						
	USI	NUSI		USI				
Collect Statistics (Y/N)								
		Y	Y					Y**

** Too volatile for stats.

Exercise 4 – Making Final Index Choices

ENTITY 1							
500,000,000 Rows	A	B	C	D	E	F	
PK/FK	PK,UA						
Value Access	250K	0	0	0	0	0	
Range Access	0	0	0	0	0	0	
Join Access	10M	0	0	0	0	0	
Join Rows	10M	0	0	0	0	0	
Distinct Values	500M	475M	1.5M	1.25M	200M	5M	
Max Rows/Value	1	2	400	450	3	110	
Max Rows/NULL	0	0	0	0	10M	0	
Typical Rows/Value	1	1	325	350	2	90	
Change Rating	0	2	1	1	1	1	
PI/SI	UPI		NUPI	NUPI		NUPI	
	USI						
Collect Statistics (Y/N)							

Exercise 4 – Making Final Index Choices (cont.)

ENTITY 2						
50,000,000 Rows	G	H	I	J	K	L
PK/FK	PK,SA					
Value Access	5K	365	52	12	0	0
Range Access	12	0	0	0	0	2.6K
Join Access	100M	0	0	0	0	0
Join Rows	100M	0	0	0	0	0
Distinct Values	50M	500K	45M	12	50	900K
Max Rows/Value	1	200	2	5M	1.2M	60
Max Rows/NULL	0	0	450K	0	0	0
Typical Rows/Value	1	100	1	4M	950K	50
Change Rating	0	0	9	1	2	0
PI/SI	UPI	NUPI				NUPI
	USI	NUSI				VONUSI
Collect Statistics (Y/N)						

Exercise 4 – Making Final Index Choices (cont.)

5,000,000 Rows	DEPENDENT						
	A	M	N	O	P	Q	
PK/FK	PK				NN,ND		
	FK	SA					
Value Access	0	0	0	0	0	0	
Range Access	0	0	0	0	0	0	
Join Access	700K	0	0	0	0	0	
Join Rows	1M	0	0	0	0	0	
Distinct Values	2M	50	90K	3M	5M	2M	
Max Rows/Value	4	200K	75	2	1	5	
Max Rows/NULL	0	0	0	390K	0	1M	
Typical Rows/Value	1	60K	50	1	1	1	
Change Rating	0	0	3	1	0	1	
PI/SI	UPI				UPI		
	NUPI						
	USI				USI		
Collect Statistics (Y/N)							

Exercise 4 – Making Final Index Choices (cont.)

ASSOCIATIVE 1					
1.5 Billion Rows	A	G	R	S	
PK/FK	PK				
	FK	FK,SA			
Value Access	26K	0	0	0	
Range Access	0	0	0	0	
Join Access	0	8M	0	0	
Join Rows	0	300M	0	0	
Distinct Values	490M	50M	75K	4M	
Max Rows/Value	5	40	21K	400	
Max Rows/NULL	0	0	0	0	
Typical Rows/Value	3	25	19K	350	
Change Rating	0	0	0	0	
PI/SI	UPI				
	NUPI	NUPI	NUPI?	NUPI	
	USI				
	NUSI				
Collect Statistics (Y/N)					

Exercise 4 – Making Final Index Choices (cont.)

ASSOCIATIVE 2						
500,000,000 Rows	A	M	G	T	U	
PK/FK	PK					
	FK		FK			
Value Access	0		0	0	0	
Range Access	0		0	0	0	
Join Access	7M		250K	0	0	
Join Rows	800M		100M	0	0	
Distinct Values	250M		50M	2.8M	750	
Max Rows/Value	3		150	180	675K	
Max Rows/NULL	0		0	0	0	
Typical Rows/Value	1		8	170	500K	
Change Rating	0		0	0	0	
PI/SI	UPI					
	NUPI		NUPI	NUPI		
	USI					
Collect Statistics (Y/N)						

What additional index choices would be available for the column G?

Exercise 4 – Making Final Index Choices (cont.)

HISTORY						
3.65 Billion Rows	A	DATE	D	E	F	
PK/FK	PK					
	FK	SA				
Value Access	10M	26K	0	0	0	
Range Access	0	100K	0	0	0	
Join Access	800M	0	0	0	0	
Join Rows	2.4B	0	0	0	0	
Distinct Values	500M	730	N/A	N/A	N/A	
Max Rows/Value	18	5.5M	N/A	N/A	N/A	
Max Rows/NULL	0	0	N/A	N/A	N/A	
Typical Rows/Value	3	4.5M	N/A	N/A	N/A	
Change Rating	0	0	N/A	N/A	N/A	
PI/SI	UPI					
	NUPI					
	USI					
	NUSI	VONUSI				
Collect Statistics (Y/N)						

What additional index choices would be available for the DATE column?

Module 22: Review Questions

1. When do Cartesian Product Joins generally occur?
 - a. When an error is made in coding the SQL query.
 - b. When a column appears in an equality join constraint.
 - c. When the join column is the primary index of both tables.
 - d. When the join column is the primary index in only one of the tables.

2. Which two types of joins will result in a Join Condition of (1=1) in EXPLAIN output? ____ ____
 - a. Hash
 - b. Merge
 - c. Nested
 - d. Exclusion merge
 - e. Cartesian product

3. Which demographic represents how often all known transactions access rows from the table through a join on this column?
 - a. Value Access Frequency
 - b. Range Access Rows
 - c. Range Access Frequency
 - d. Join Access Rows
 - e. Join Access Frequency

Module 23: Review Questions

Check the box if the attribute applies to the index.

	Compressed Join Index Syntax	Non- Compressed Join Index Syntax	Aggregate Join Index	Sparse Join Index	Hash Index
May be created on a single table					
May be created on multiple tables					
Requires the use of SUM, COUNT or MIN/MAX functions					
Requires a WHERE condition to limit rows stored in the index					
Automatically updated as base table rows are inserted or updated					
Automatically includes the base table PI value as part of the index					

Module 24: Review Questions

1. Which BTEQ setting controls Teradata vs. ANSI mode? _____
2. Which commands will not work in ANSI mode? _____
3. True or False. A syntax error for an individual request in ANSI (or COMMIT) mode causes the entire transaction to be rolled back.
4. True or False. A syntax error for an individual request in Teradata (or BTET) mode causes the entire transaction to be rolled back.
5. True or False. Logging off during an explicit transaction without either a COMMIT or ET will always result in a ROLLBACK.
6. Where does a Volatile Temporary table get its space from? _____
7. Where does a Global Temporary table get its space from? _____

Module 25: Review Questions

Answer True or False.

1. True or False. With TPT Update, you can import and export data.
2. True or False. In Teradata mode, an implicit DELETE ALL transaction does not use the Transient Journal to store before-images of deleted rows.
3. True or False. An INSERT/SELECT of 1,000,000 rows into an empty table is only slightly faster than an INSERT/SELECT of 1,000,000 rows into a table with 1 row.

Match the Teradata Parallel Transporter operator with the corresponding Teradata utility.

- | | |
|---------------|---------------|
| 1. ___ UPDATE | A. MultiLoad |
| 2. ___ STREAM | B. FastLoad |
| 3. ___ LOAD | C. FastExport |
| 4. ___ EXPORT | D. TPump |

Appendix A: The Lab Environment

**This appendix describes the lab environment that
will be used during the class.**

Teradata Proprietary and Confidential

Lab Environment

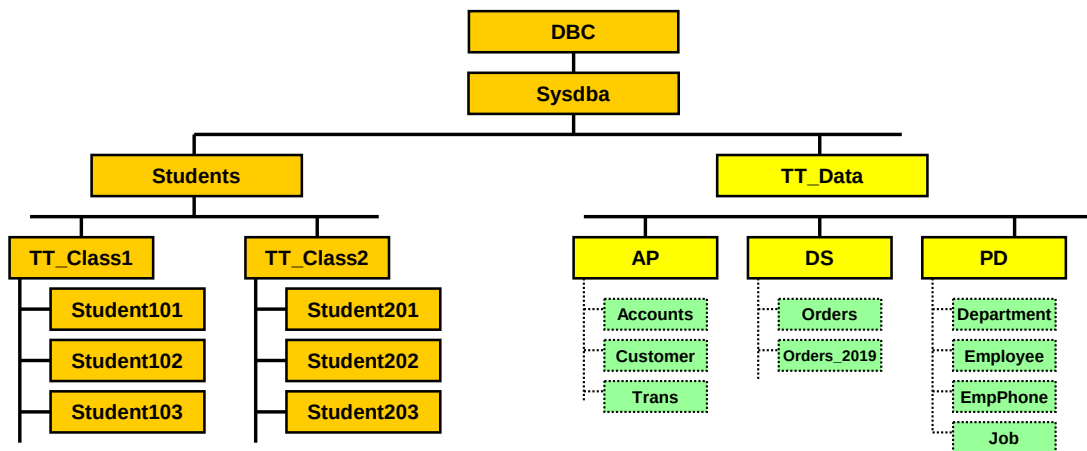
Most access to Teradata will be through Teradata SQL Assistant and an ODBC connection.

However, you may need to connect from a Windows PC to a Linux system. Linux requires a Secure Shell connection. This opens a terminal window which you can use to logon to Linux and enter Linux commands.

putty is an example of a utility that provides a secure shell connection

If you are executing BTEQ interactively, you cannot enter your password as part of the .LOGON statement. BTEQ will prompt you for your Teradata username password.

Lab Environment



Common Teradata SQL Commands:

To populate a newtable:

```
INSERT INTO newtable
SELECT * FROM db.olddtab WITH [NO] DATA;
```

HELP USER username;	- lists objects owned by user
HELP 'SQL sqlcommand';	- help for an SQL command
SHOW TABLE tablename;	- shows table creation syntax
SELECT COUNT(*) FROM tname;	- provides count of rows
DELETE tablename ALL;	- deletes all rows in a table
DROP TABLE tablename;	- drops table from database

Useful Linux Commands:

ls -l	- long listing of Linux files)
more filename	- view contents of text file
vi filename	- create or edit a text file
cp file1 file2	- copies file1 to file2
mv oldname newname	- renames a file
rm filename	- delete a file
exit	- exits Linux

AP, DS, and PD Databases

The following three pages list the key tables in the AP, DS, and PD databases that will be used throughout this course.

AP Tables

AP.Accounts (10,000 Rows)

Account Number	Street Number	Street	City	State	Zip Code	Balance Forward	Balance Current
PK							
UPI							
20024010	123	Harbor Blvd.	Torrance	CA	90323	1000.00	900.00
20031023	3456	186th St.	Glendale	CA	90451	1500.00	1700.00
20049873	100	Western Av.	Las Vegas	NV	97345	400.00	400.00
20031134	10	Heather Rd.	S. Monica	CA	92345	6020.00	5312.00

Integer Integer Char(25) Char(20) Char(2) Integer Decimal(10,2) Decimal(10,2)

Note:

Column names that span 2 lines are connected with an underscore.

Ex: Account_Number

AP.Customer (7,000 Rows)

Customer Number	Last Name	First Name	Social Security
PK			
UPI			
13021	Smith	George	456788765
18765	Jones	Barbara	987453498
11023	Wilson	John	495028367
1123	Omaguchi	Sandra	234904587

Integer Char(30) Char(20) Integer

AP.Trans (15,000 Rows)

Trans Number	Trans Date	Account Number	Trans ID	Trans Amount
PK				
		NUPI		
4653	2002-02-11	20024020	2009	-50.00
3241	2002-02-08	20034567	DEP	160.00
1298	2002-02-08	20024005	2987	-70.00
11026	2002-02-13	20024020	DEP	20.00

Integer Date Integer Char(4) Decimal(10,2)

DS Tables

Orders (350,000 Rows)

orderid	custid	orderstatus	ordertotal	orderdate	orderpriority	clerk	location	shippriority	ordercomment
PK	FK,NN								
UPI									
100001	1001	C	1,005.00	2008-01-02	10	Jack ..	5	20	In Stock
103501	1451	C	1,005.00	2009-12-01	10	Dee ...	3	20	In Stock
101400	1080	C	1,150.00	2010-02-28	10	Fred ...	10	20	In Stock
Integer	Integer	CHAR(1)	Decimal (10,2)	Date	SmallInt	Char(16)	SmallInt	SmallInt	Varchar(79)

Orders_2019 (100,000 Rows) – Same Layout

PD Tables

Employee (150,000 Rows)

Emp Number	Dept Number	EmpMgr Number	Job Code	Last Name	First Name	Salary Amount
PK	FK	FK	FK			
UPI						
100001	999	?	8903	DeBosse	Ibee	200000.00
100797	1048	100791	3017	Myers	Ruth	41000.00
100002	1001	100001	3001	Smith	Steve	110000.00
Integer	Integer	Integer	Integer	Char(20)	Varchar(20)	Decimal (10,2)

Job (11,124 Rows)

Job Code	Job Desc
PK	
UPI	
3000	President
3001	Senior Mgmt
3017	Analyst L2
Integer	Char(50)

Department (17,415 Rows)

Dept Number	Dept Name	DeptMgr Number	Budget Amount
PK		FK	
UPI			
1048	Design SW	100791	1000000.00
1050	Design Services	100811	1000000.00
1028	Engineering SW	100441	3000000.00
Integer	Char(100)	Integer	Decimal (10,2)

EmpPhone (350,000 Rows)

Emp Number	Area Code	Phone Number	Extension
PK			
FK			
NUPI			
100001	937	5100001	1001
100001	937	4100001	1001
100389	858	4852815	815
Integer	SmallInt	Integer	Integer

Note:

Column names that span 2 lines are NOT connected with an underscore for these tables.
Ex: EmpNumber

vi or vim Notes

You can use vi or vim to build your scripts. It may be easier to create your script in a Windows Notepad file, copy it, and paste it into a vi file.

Another technique that can be used to create Linux scripts **without** using vi is to do the following:

1. Enter your commands (job/script) in a Notepad file.
2. Highlight the text and use the mouse to choose the **Edit → Copy** function

Switch to your terminal window where Linux is running and ...

3. **cat > labx_1.btq** (or whatever filename you wish)

Use the mouse to choose the **Edit → Paste** function

To exit the cat command, press either the DELETE key or CNTL C.

vi or vim Commands

vim filename

Movement and other

h move cursor to left
j move cursor down a line
k move cursor up a line
l move cursor to right
space move cursor to right
w move a word to right
b move a word to left
0 go to start of a line (zero)
\$ go to end of a line
G go to end of file
nG go to line n
^f scroll forward 1 screen
^b scroll back 1 screen
^l refresh the screen
/str search forward for str
· repeat last command

Modification

r replace 1 character
x delete char under cursor
X delete char before cursor
5x delete 5 characters
dw delete a word
dd delete the line
d\$ delete the rest of the line
dG delete rest of the file
cw change a word
C change rest of the line
J join 2 lines together
yy yank or copy a line
3yy yank or copy 3 lines
p put yanked line(s) after line
P put yanked line(s) before line
u undo last command

Input mode

i insert before cursor
I insert at beginning of line
a insert after cursor
A insert at end of line
o open a new line below
O open a new line above
R Replace mode and insert after <cr>

ESC exits
input &
colon
modes

Colon mode

:w write or save the file
:w fname write new file name
:wq write and quit
:q! quit and don't save

:e fname edit another file
:r fname read in a file
:!cmd run UNIX command
:set smd set showmode on

Notes

Lab Exercises

**This section includes the lab exercises for the
Teradata Factory Xpress course.**

Teradata Proprietary and Confidential

Lab Exercise 7-1

Check your understanding of the concepts discussed in this module by completing the lab exercise as directed by your instructor.

Lab Exercise 7-1

Purpose

In this lab, you will use either Teradata Studio or Teradata Studio Express to define a data source and execute some simple SQL commands.

What you need

Teradata Studio or Teradata Studio Express installed on the laptop or PC.

Tasks

1. If this has not been already defined, create a connection to a Teradata system based on the following:

Complete the dialog box with the following information:

Name – Factory

Description – Teradata Training for your name

Name or IP Address – _____ (supplied by instructor)

Username – _____ (supplied by instructor)

Password – do not fill in your password

Verify the following options are set properly.

Session Character Set – ASCII

Session Mode – Teradata (TMODE value of TERA)

Lab Exercise 7-1 (cont.)

Check your understanding of the concepts discussed in this module by completing the lab exercise as directed by your instructor.

Lab Exercise 7-1

2. Connect to the data source you just created and logon with your username and password.
3. Use the Window > Preferences > Teradata Datatools Preferences tab, ensure the following options are set as indicated:
 - Data Source Explorer Load Preferences
 - Check** – Only Databases (but include DBC and myself)
 - Un-Check** – Show Databases and Users in Hierarchical Display
 - Result Set Viewer Preferences
 - Check** – Display alternate Result Set rows in color – choose a color
 - Check** – Display Column Titles rather than Column Names
4. If the Explorer Tree pane is not visible, mouse click > next to your login name and then Databases
 - Find the following databases in the Explorer Tree: AP, DS, PD, Collaterals
5. Using the Explorer Tree, view the table objects in your database.

Lab Exercise 7-1 (cont.)

Check your understanding of the concepts discussed in this module by completing the lab exercise as directed by your instructor.

Use the following SQL to determine a count of rows in a table.

```
SELECT COUNT(*) FROM tablename;
```

Step 7 Hint: Your Orders_1Q2019 table should have 8000 rows for just January 2019 orders. If not, check the dates you used in your queries.

Step 8 Hint: Your Orders_1Q2019 table should have 24,000 rows for orders in the first quarter of 2019. If not, check the dates you used in your queries.

Lab Exercise 7-1 (cont.)

6. Using the Query Window, execute the following query.

```
CREATE TABLE Orders_1Q2019 AS Orders_2019 WITH NO DATA;
```

Does the new table object appear in the table object list? ____ If not, "refresh" the database.

7. Execute the following query to populate Orders_1Q2019 with orders for January of 2019.

```
INSERT INTO Orders_1Q2019 SELECT * FROM DS.Orders_2019  
WHERE orderdate BETWEEN DATE '2019-01-01' AND DATE '2019-01-31';
```

How many rows are in Orders_1Q2019? ____

8. Using the History window, recall the query from step #7 and modify it to add orders for February and March of 2019 into Orders_1Q2019.

How many rows are now in Orders_1Q2019? ____

9. Execute the following query by using the drag and drop object feature of the SQL tool you are using.

```
SELECT custid, SUM (ordertotal)  
FROM Orders_1Q2019  
GROUP BY 1  
ORDER BY 1;
```

What is the SUM of the orders (ordertotal) in the 1Q2019 for custid = 21012? ____

Lab Exercise 7-1 (cont.)

Check your understanding of the concepts discussed in this module by completing the lab exercise as directed by your instructor.

Use the following SQL to create a view.

```
CREATE VIEW viewname  
AS  
LOCKING ROW FOR ACCESS  
SELECT      column1, column2  
FROM        table_or_view_name  
WHERE       condition;
```

To create a simple macro (without parameters), you can use the following SQL:

```
CREATE      MACRO macroname  
AS         (SELECT * FROM table_or_view_name);;
```

Use the following SQL to execute a simple macro.

```
EXEC macroname;
```

To create a parameterized macro, you can use the following SQL:

```
CREATE      MACRO orderid_m (oid_value INTEGER)  
AS         (SELECT * FROM table_or_view_name  
            WHERE Orderid = :oid_value);;
```

Use the following SQL to execute a parameterized macro.

```
EXEC orderid_m (1781509);
```

```
EXEC orderid_m (1781939);
```

Lab Exercise 7-1 (cont.)

10. Create a view (that utilizes an Access Lock) named "Orders_Jan2019_v" on the Orders_1Q2019 table that includes the following columns and only includes orders for the January of 2019.

orderid, custid, ordertotal, orderdate

SELECT the total count of orders from "Orders_Jan2019_v".

How many orders are available using this view? _____

11. Create a parameterized macro named "Orderid_m" which accepts an orderid as a value, accesses the view, and returns a single order from "Orders_Jan2019_v".

Execute the macro to provide order details for orders 1781509 and 1781939.

What is the ordertotal for orderid #1781509 _____

What is the ordertotal for orderid #1781939? _____

12. (Optional) Use the Collaterals database to access the Photos table to display various JPG files.

Execute the following: `SELECT * FROM Collaterals.Photos ORDER BY 1;`

Note: Set the file type to JPG and check the option "Display as picture in Answerset".

What is shown in photo #7? _____

Lab Exercise 11-1

Check your understanding of the concepts discussed in this module by completing the lab exercise as directed by your instructor.

The SQL to populate and verify the row count of your Employee table is:

```
INSERT INTO Employee SELECT * FROM PD.Employee;  
SELECT COUNT(*) FROM Employee;
```

Lab Exercise 11-1

Lab Exercise 11-1

Purpose

In this lab, you will use Teradata Studio or Teradata Studio Express to evaluate various columns of table as primary index candidates.

What you need

Populated PD.Employee table; your empty Employee table

Tasks

1. Using INSERT/SELECT, populate your "Employee" table with all rows from the PD.Employee table. Verify the number of rows in your table.

```
INSERT INTO Employee SELECT * FROM PD.Employee;
```

```
SELECT COUNT(*) FROM Employee;    Count = _____
```

Lab Exercise 11-1 (cont.)

Use the following SQL to determine the column metrics for this Lab.

of Distinct Values for a column:

```
SELECT    COUNT(DISTINCT(column_name))
FROM      tablename;
```

Max Rows per Value for all values in a column:

```
SELECT    column_name, COUNT(*)
FROM      table_name
GROUP BY  1
ORDER BY  2 DESC;
```

Max Rows per Value for Top 3 values in a column:

```
SELECT    TOP 3 tvalue, tcount
FROM      (SELECT    column_name, COUNT(*)
           FROM      table_name GROUP BY 1)
           t1 (tvalue, tcount)
ORDER BY  tcount DESC;
```

Max Rows with NULL in a column:

```
SELECT    COUNT(*)
FROM      table_name
WHERE     column_name IS NULL;
```

Average Rows per Value for a column:

```
SELECT    COUNT(*)/COUNT(DISTINCT(column_name)) AS averagects
FROM      table_name;
```

Typical (Median) Rows per Value for a column:

```
SELECT    MEDIAN(cts)AS mediancts
FROM      (SELECT    column_name,
                   COUNT(*) AS cts,
                   COUNT(cts) OVER (PARTITION BY cts) AS ctscs
           FROM      table_name
           GROUP BY 1) t1;
```

Lab Exercise 11-1 (cont.)

2. Collect column demographics for each of these columns in Employee and determine if the column would be a primary index candidate or not.

By using the SHOW TABLE Employee command, you should be able to complete the EmpNumber information without executing any SQL.

	Distinct Values	Max Rows for a Value	Max Rows NULL	Avg Rows per Value	Candidate for PI (Y/N)
EmpNumber					
DeptNumber					
JobCode					
LastName					

Lab Exercise 11-2

Distribution of table space by AMP:

```
SELECT      Vproc, TableName (CHAR(15)), CurrentPerm
FROM        DBC.TableSizeV
WHERE       DatabaseName = DATABASE
AND         TableName = 'table_name'
ORDER BY    1;
```

To determine the minimum and maximum space used by AMP:

```
SELECT      MIN(CurrentPerm) AS Min_Size,
            MAX(CurrentPerm) AS Max_Size
FROM        DBC.TableSizeV
WHERE       DatabaseName = DATABASE
AND         TableName = 'table_name';
```

To display sum of space, minimum space used by an AMP, maximum space used by an AMP in MB (Megabytes) AND calculate a Skew Factor for all the tables in your database:

```
SELECT      TableName,
            SUM(CurrentPerm)/(1024**2) AS CurentPerm_MB,
            MIN(CurrentPerm)/(1024**2) AS Min_Size,
            MAX(CurrentPerm)/(1024**2) AS Max_Size,
            (100 - (AVG(CurrentPerm)/
                    NULLIFZERO (MAX(CurrentPerm))* 100))
                                                    AS SKEW_FACTOR
FROM        DBC.TableSizeV
WHERE       DatabaseName = DATABASE
GROUP BY    1
ORDER BY    1;
```

As an alternative to display this information in MB (Megabytes) AND calculate a Skew Factor for all the tables in your database, you can also use the following:

```
SELECT      TableName,
            SUM(CurrentPerm)/(2**20) AS CurentPerm_MB,
            MIN(CurrentPerm)/(2**20) AS Min_Size,
            MAX(CurrentPerm)/(2**20) AS Max_Size,
            (100 - (AVG(CurrentPerm)/
                    NULLIFZERO (MAX(CurrentPerm))* 100))
                                                    AS SKEW_FACTOR
FROM        DBC.TableSizeV
WHERE       DatabaseName = DATABASE
GROUP BY    1
ORDER BY    1;
```

Lab Exercise 11-2

Lab Exercise 11-2

Purpose

In this lab, you will use the DBC.TableSizeV view to determine space distribution on a per AMP basis.

What you need

Your populated Employee table.

Tasks

1. Use SHOW TABLE command to determine which column is the Primary Index. PI = _____

Determine the AMP space usage of your Employee table using DBC.TableSizeV.

What is the least amount of permanent space on one of the AMPs? size _____

What is the greatest amount of permanent space on one of the AMPs? size _____

2. Create a new table named Employee_2 with the same columns as Employee except specify LastName as the Primary Index.

Using INSERT/SELECT, populate your Employee_2 table with all rows from your Employee table.

Determine the AMP space usage of your Employee_2 table using DBC.TableSizeV.

What is the least amount of permanent space on one of the AMPs? size _____

What is the greatest amount of permanent space on one of the AMPs? size _____

Lab Exercise 11-3 (optional)

Distribution of table space by AMP:

```
SELECT      Vproc, TableName (CHAR(15)), CurrentPerm
FROM        DBC.TableSizeV
WHERE       DatabaseName = DATABASE
AND         TableName = 'table_name'
ORDER BY    1;
```

To determine the minimum and maximum space used by AMP:

```
SELECT      MIN(CurrentPerm) AS Min_Size,
            MAX(CurrentPerm) AS Max_Size
FROM        DBC.TableSizeV
WHERE       DatabaseName = DATABASE
AND         TableName = 'table_name';
```

To display sum of space, minimum space used by an AMP, maximum space used by an AMP in MB (Megabytes) AND calculate a Skew Factor for all the tables in your database:

```
SELECT      TableName,
            SUM(CurrentPerm)/(1024**2) AS CurentPerm_MB,
            MIN(CurrentPerm)/(1024**2) AS Min_Size,
            MAX(CurrentPerm)/(1024**2) AS Max_Size,
            (100 - (AVG(CurrentPerm)/
                    NULLIFZERO (MAX(CurrentPerm))* 100))
                                                    AS SKEW_FACTOR
FROM        DBC.TableSizeV
WHERE       DatabaseName = DATABASE
GROUP BY    1
ORDER BY    1;
```

As an alternative to display this information in MB (Megabytes) AND calculate a Skew Factor for all the tables in your database, you can also use the following:

```
SELECT      TableName,
            SUM(CurrentPerm)/(2**20) AS CurentPerm_MB,
            MIN(CurrentPerm)/(2**20) AS Min_Size,
            MAX(CurrentPerm)/(2**20) AS Max_Size,
            (100 - (AVG(CurrentPerm)/
                    NULLIFZERO (MAX(CurrentPerm))* 100))
                                                    AS SKEW_FACTOR
FROM        DBC.TableSizeV
WHERE       DatabaseName = DATABASE
GROUP BY    1
ORDER BY    1;
```

Lab Exercise 11-3 (optional)

Lab Exercise 11-3 (Optional)

Purpose

In this lab, you will determine data distribution on a per AMP basis for NoPI tables.

What you need

Your populated Employee_2 table (created in the previous exercise).

Tasks

1. Use the SHOW TABLE for Employee_2 to help create a new, similar table (same column names and definitions, etc.) named "Employee_NoPI" and does NOT have a primary index.

Using INSERT/SELECT, populate the Employee_NoPI table with all rows from your Employee_2 table.

Determine the AMP space usage of the Employee_NoPI table using DBC.TableSizeV.

What is the least amount of permanent space on one of the AMPs? size _____

What is the greatest amount of permanent space on one of the AMPs? size _____

Is the distribution similar to the Employee_2 table (yes or no)? _____

Why or why not? _____

Lab Exercise 11-3 (optional) – cont.

Distribution of table space by AMP:

```
SELECT      Vproc, TableName (CHAR(15)), CurrentPerm
FROM        DBC.TableSizeV
WHERE       DatabaseName = DATABASE
AND         TableName = 'table_name'
ORDER BY    1;
```

To determine the minimum and maximum space used by AMP:

```
SELECT      MIN(CurrentPerm) AS Min_Size,
            MAX(CurrentPerm) AS Max_Size
FROM        DBC.TableSizeV
WHERE       DatabaseName = DATABASE
AND         TableName = 'table_name';
```

To display sum of space, minimum space used by an AMP, maximum space used by an AMP in MB (Megabytes) AND calculate a Skew Factor for all the tables in your database:

```
SELECT      TableName,
            SUM(CurrentPerm)/(1024**2) AS CurentPerm_MB,
            MIN(CurrentPerm)/(1024**2) AS Min_Size,
            MAX(CurrentPerm)/(1024**2) AS Max_Size,
            (100 - (AVG(CurrentPerm)/
                    NULLIFZERO (MAX(CurrentPerm))* 100))
                                AS SKEW_FACTOR
FROM        DBC.TableSizeV
WHERE       DatabaseName = DATABASE
GROUP BY    1
ORDER BY    1;
```

As an alternative to display this information in MB (Megabytes) AND calculate a Skew Factor for all the tables in your database, you can also use the following:

```
SELECT      TableName,
            SUM(CurrentPerm)/(2**20) AS CurentPerm_MB,
            MIN(CurrentPerm)/(2**20) AS Min_Size,
            MAX(CurrentPerm)/(2**20) AS Max_Size,
            (100 - (AVG(CurrentPerm)/
                    NULLIFZERO (MAX(CurrentPerm))* 100))
                                AS SKEW_FACTOR
FROM        DBC.TableSizeV
WHERE       DatabaseName = DATABASE
GROUP BY    1
ORDER BY    1;
```

Lab Exercise 11-3 (optional) – cont.

2. DELETE all the rows in Employee_NoPI.

Using INSERT/SELECT with the **HASH BY RANDOM** option, populate your **Employee_NoPI** with all rows from your **Employee_2** table.

Determine the AMP space usage of your **Employee_NoPI** table using DBC.TableSizeV.

What is the least amount of permanent space on one of the AMPs? size _____

What is the greatest amount of permanent space on one of the AMPs? size _____

Is the space distribution the same as the **Employee_2** table (yes or no)? _____

Is the table distributed as evenly as expected (yes or no)? _____

Why or why not? _____

3. DELETE all the rows in Employee_NoPI.

Using INSERT/SELECT with the **HASH BY RANDOM (1, 2000000000)** option, populate **Employee_NoPI** with all the rows from your **Employee_2** table.

Determine the AMP space usage of your **Employee_NoPI** table using DBC.TableSizeV.

What is the least amount of permanent space on one of the AMPs? size _____

What is the greatest amount of permanent space on one of the AMPs? size _____

Is the space distribution similar to the **Employee** table (yes or no)? _____

Is the table distributed as evenly as expected (yes or no)? _____

Why or why not? _____

Lab Exercise 14-1

Check your understanding of the concepts discussed in this module by completing the lab exercises as directed by your instructor.

SQL hints:

```
INSERT INTO table_1 SELECT * FROM table_2;
```

```
SELECT COUNT(*) FROM table_name;
```

```
SHOW TABLE table_name;
```

A count of rows in the Orders table is 350,000.

A count of rows in the Orders_2019 table is 100,000.

Lab Exercise 14-1

Lab Exercise 14-1

Purpose

In this lab, you will create tables with primary indexes partitioned in various ways.

What you need

Populated DS tables and empty tables in your database

Tasks

1. Using INSERT/SELECT, populate your Orders and Orders_2019 tables from the DS.Orders and DS.Orders_2019 tables, respectively. Your Orders table will have data for the years 2010 to 2018 and the Orders_2019 table will have data for 2019. Verify the number of rows in your tables.

```
SELECT COUNT(*) FROM Orders;          Count = _____ (should be 350,000)
SELECT COUNT(*) FROM Orders_2019;     Count = _____ (should be 100,000)
```

2. Use the SHOW TABLE for Orders to help create a new, similar table (same column names and definitions, etc.) named "Orders_RP" that is row partitioned based on the following:

Primary Index – orderid

Partitioning column – orderdate

- From '2010-01-01' through '2019-12-31', partition by month
- Include the NO RANGE option (the UNKNOWN option is not needed for orderdate)
- Do not create any secondary indexes for this table

How many partitions are logically defined for the Orders_RP table? _____

Lab Exercise 14-1 (cont.)

Check your understanding of the concepts discussed in this module by completing the lab exercises as directed by your instructor.

SQL hints:

```
INSERT INTO table_1 SELECT * FROM table_2;
```

```
SELECT      COUNT(*)  
FROM        table_name;
```

```
SELECT      PARTITION, COUNT(*)  
FROM        table_name  
GROUP BY    1  
ORDER BY    1;
```

To determine the number of partitions in a table.

```
SELECT      COUNT(DISTINCT(PARTITION)) FROM table_name;
```

To determine the number of orders (rows) for months in a specific year.

```
SELECT      PARTITION, COUNT(*)  
FROM        table_name  
WHERE       EXTRACT(YEAR FROM date_column_name) = 2019  
AND         EXTRACT(MONTH FROM date_column_name) between 1 and 12  
GROUP BY    1  
ORDER BY    1;
```

To determine the number of orders (rows) for a specific month:

```
SELECT      COUNT(*)  
FROM        table_name  
WHERE       EXTRACT(YEAR FROM date_column_name) = 2019  
AND         EXTRACT(MONTH FROM date_column_name) = 1;
```

Lab Exercise 14-1 (cont.)

3. INSERT/SELECT all of the rows from the Orders table into the Orders_RP table. Verify the number of rows in your table. Count = _____

How many partitions would you estimate have data at this time? _____

4. Use the PARTITION key word to list the partitions and number of rows in various partitions.

How many partitions actually have data? _____

How many rows are in the partition for January, 2010? _____

How many rows are in the partition for December, 2018? _____

5. Use INSERT/SELECT to add the rows from the Orders_2019 table to your Orders_RP table. Verify the number of rows in your table. Count = _____

Use the PARTITION key word to determine the number of partitions used and the number of rows in various partitions.

How many partitions actually have data? _____

How many rows are in the partition for the January, 2019? _____

Lab Exercise 14-1 (cont.)

Check your understanding of the concepts discussed in this module by completing the lab exercises as directed by your instructor.

SQL hints:

```
INSERT INTO table_1 SELECT * FROM table_2;
```

```
SELECT      COUNT(*) FROM table_name;
```

To determine the number of partitions in a table.

```
SELECT      COUNT(DISTINCT(PARTITION)) FROM table_name;
```

To determine the number of orders (rows) for a specific month:

```
SELECT      COUNT(*)  
FROM        table_name  
WHERE       EXTRACT(YEAR FROM date_column_name) = 2019  
AND         EXTRACT(MONTH FROM date_column_name) = 1;
```

Other SQL queries that may be useful include:

```
SELECT      COUNT(DISTINCT(PARTITION))  
FROM        table_name  
WHERE       orderdate ... ;
```

```
SELECT      MAX(PARTITION)  
FROM        table_name;
```

```
SELECT      PARTITION, COUNT(*)  
FROM        table_name  
GROUP BY    1  
ORDER BY    1;
```

The PARTITION key word only returns partition numbers of partitions that contain rows. The following “canned” SQL can be used to return a list of partitions that are not used between the first and last used partitions.

```
SELECT p + 1      AS "The missing partitions are:"  
FROM  
  (SELECT          p1 - p2      AS p,  
                   PARTITION AS p1,  
                   MDIFF(PARTITION, 1, PARTITION) AS p2  
   FROM            table_name  
   QUALIFY         p2 > 1)  
AS temp;
```

Lab Exercise 14-1 (cont.)

6. INSERT the following 2 rows (using these values) into the Orders_RP table.

(50000, 1000, 'C', 1000, '2020-01-01', 10, 'your name', 5, 20, 'order for January in 2020')
 (50001, 1001, 'C', 1001, '2020-02-01', 10, 'EF Codd', 5, 20, 'order for February in 2020')

How many partitions are now in Orders_RP? ____

What is the partition number (highest partition #) of the NO RANGE partition? ____

7. Create a new table named "Orders_RP_ML" that has multilevel row partitioning based on the following:

Primary Index – orderid

First Level of Partitioning column – orderdate (use month ranges for all 10 years)

Include the NO RANGE option for orderdate

Second Level of Partitioning column – location (10 different order locations (1 through 10)

Place the NO RANGE and UNKNOWN rows into the same special partition for location

Unique secondary index – orderid

8. Populate the Orders_RP_ML table from the Orders and Orders_2019 tables using INSERT/SELECT and execute these commands in parallel. Verify the number of rows in Orders_RP_ML.
 Count = _____

Lab Exercise 14-1 (cont.)

Check your understanding of the concepts discussed in this module by completing the lab exercises as directed by your instructor.

SQL hints:

```
INSERT INTO table_1 VALUES (value1, value2, ...);
```

```
INSERT INTO table_1 SELECT * FROM table_2;
```

```
SELECT COUNT(*) FROM table_name;
```

To determine the number of partitions in a table.

```
SELECT      COUNT(DISTINCT(PARTITION)) FROM table_name;
```

To determine the number of orders (rows) for a specific month:

```
SELECT      COUNT(*)  
FROM        table_name  
WHERE       EXTRACT(YEAR FROM date_column_name) = 2019  
AND         EXTRACT(MONTH FROM date_column_name) = 1;
```

Other SQL queries that may be useful include:

```
SELECT      COUNT(DISTINCT(PARTITION))  
FROM        table_name  
WHERE       orderdate ... ;
```

```
SELECT      MAX(PARTITION)  
FROM        table_name;
```

```
SELECT      PARTITION, COUNT(*)  
FROM        table_name  
GROUP BY    1  
ORDER BY    1;
```

Lab Exercise 14-1 (cont.)

9. For the Orders_RP_ML table, use the PARTITION key word to answer the following questions.

How many partitions actually have data? _____

What is the highest partition number? _____

What is the partition number for orders in January, 2019 and location 1? _____

What is the partition number for orders in February, 2019 and location 1? _____

Is there a difference of 11 partitions between these two months? _____

Why or why not? _____

10. Before altering the table, verify the number of rows in Orders_RP. Count = _____

Use the ALTER TABLE command on Orders_RP to do the following.

- DROP RANGE (with DELETE) for year 2010
- ADD RANGE for orders that will be placed in year 2020 with an interval of 1 month

Use SHOW TABLE on Orders_RP to view the altered partitioning.

Use the PARTITION key word to list the partitions and the number of rows in various partitions.

How many partitions currently have data rows? _____

How many rows now exist in the table? _____ Has the row count changed? ____

If the row count did not change, why not? _____

How many rows are in the partition for January, 2020? _____

How many rows are in the partition for February, 2020? _____

How many rows are in the NO RANGE partition? _____

Lab Exercise 15-1

Check your understanding of the concepts discussed in this module by completing the lab exercises as directed by your instructor.

SQL hints:

```
INSERT INTO table_1 SELECT * FROM table_2;
```

```
SELECT COUNT(*) FROM table_name;
```

```
SHOW TABLE table_name;
```

A count of rows in the Orders table is 350,000.

A count of rows in the Orders_2019 table is 100,000.

Lab Exercise 15-1

Lab Exercise 15-1

Purpose

In this lab, you will create tables with column partitioning in various ways. These tables will be used in later labs.

What you need

Populated DS tables and Orders and Orders_2019 tables in your database

Tasks

1. Use the SHOW TABLE for Orders to help create a new, similar table (same column names and definitions, etc.) that does NOT have a primary index and name this table "Orders_NoPI".
2. Populate the Orders_NoPI table (via INSERT/SELECT) with all of the rows from the DS.Orders and DS.Orders_2019 tables.

Verify the number of rows in your table. Count = _____ (count should be 450,000)

3. Use the SHOW TABLE for Orders_NoPI to create a new column partitioned table named "Orders_CP" based on the following:
 - Each column is created as a separate partition
 - Utilize auto compression for every column

Populate the Orders_CP table (via INSERT/SELECT) from the Orders_NoPI table.

Lab Exercise 15-1 (cont.)

Check your understanding of the concepts discussed in this module by completing the lab exercises as directed by your instructor.

SQL hints:

```
INSERT INTO table_1 SELECT * FROM table_2;
```

```
SELECT COUNT(*) FROM table_name;
```

```
SELECT COUNT(DISTINCT(PARTITION)) FROM table_name;
```

Lab Exercise 15-1 (cont.)

4. Verify the number of rows in your table. Count = _____ (count should be 450,000)

How many partitions actually have data? _____

Note: The table only has 1 logical partition.

5. Use the SHOW TABLE for Orders_CP to create a new column partitioned table named "Orders_CP_noAC" based on the following:

- Each column is created as a separate partition
- Turn off auto compression for every column

Populate the Orders_CP_noAC table (via INSERT/SELECT) from the Orders_NoPI table.

6. Use the SHOW TABLE for Orders_CP to create a new column partitioned table named "Orders_CPPA" based on the following:

- Primary AMP Index using **custid** column and each column is created as a separate partition
- Utilize auto compression for every column

Populate the Orders_CPPA table (via INSERT/SELECT) with all of the rows from your Orders_NoPI table.

Lab Exercise 15-1 (cont.)

Check your understanding of the concepts discussed in this module by completing the lab exercises as directed by your instructor.

SQL hints:

```
SELECT      TableName, SUM(CurrentPerm)
FROM        DBC.TableSizeV
WHERE       DatabaseName = DATABASE
AND         TableName In ('tablename1', 'tablename2', ...)
GROUP BY    1
ORDER BY    1;
```

```
INSERT INTO table_1 SELECT * FROM table_2;
```

```
SELECT COUNT(*) FROM table_name;
```

Lab Exercise 15-1 (cont.)

7. Use the SHOW TABLE for Orders_CPPA to create a new column partitioned table named "Orders_CPPI" based on the following:
 - Primary Index using **custid** column and each column is created as a separate partition
 - Utilize auto compression for every column

Populate the Orders_CPPI table (via INSERT/SELECT) with all of the rows from the DS.Orders and DS.Orders_2018 tables in parallel.

Note: To improve insert performance run both insert/selects in parallel (Execute Parallel - F9 in SQL Assistant or Execute All - Ctrl+Alt+x In Teradata Studio Express)

8. Determine the AMP space usage of the Orders_CP, Orders_CP_noAC, Orders_CPPA and Orders_CPPI tables using the DBC.TableSizeV.

CurrentPerm for Orders_CP	_____
CurrentPerm for Orders_CP_noAC	_____
CurrentPerm for Orders_CPPA	_____
CurrentPerm for Orders_CPPI	_____

Which table is the largest and why? _____

What is the probable reason that the CPPA and CPPI tables use less perm space?

Lab Exercise 15-1 (cont.)

Check your understanding of the concepts discussed in this module by completing the lab exercises as directed by your instructor.

SQL hints:

```
INSERT INTO table_1 SELECT * FROM table_2;
```

```
SELECT COUNT(*) FROM table_name;
```

```
SELECT COUNT(DISTINCT(PARTITION)) FROM table_name;
```

Lab Exercise 15-1 (cont.)

9. Use SHOW TABLE for Orders_CP to create a new column partitioned table named "Orders_CP_RP" based on the following:

- Incorporate row partitioning with orderdate as the partitioning column
 - From '2010-01-01' through '2019-12-31', partition by day
 - Do not use the NO RANGE or UNKNOWN options.
- Each column is created as a separate partition (COLUMN partitioning is the second level)
- Utilize auto compression for every column

Populate the Orders_CP_RP table (via INSERT/SELECT) from the Orders_NoPI table.

10. Use the PARTITION key word to determine the number of partitions defined in the Orders_CP_RP.

How many partitions actually have data? _____

Lab Exercise 18-1

The following SQL can be used to determine the size of a table.

SUM of Perm space using the DBC.TableSizeV view.

```
SELECT      TableName (CHAR(15)), SUM(CurrentPerm)
FROM        DBC.TableSizeV
WHERE       DatabaseName = DATABASE
AND         TableName = 'tablename'
GROUP BY    1
ORDER BY    1;
```

Remember that compression is case-sensitive. The clerk names have to be compressed as shown on the facing page OR you can paste the clerk names from below into the COMPRESS option on the clerk column of the CREATE TABLE DDL statement.

```
'April Rains', 'Auggie Sunshine', 'Dee Winter', 'Fred Cold',
'Jack Snow', 'June Summer', 'Justin Heat', 'Matt Winds',
'May Flowers', 'Norbert Thanks', 'Ossie Leaves', 'Steve Fall'
```

Lab Exercise 18-1

Lab Exercise 18-1

Purpose

In this lab, you will compress multiple values for a column in order to reduce Perm space.

What you need

Populated Orders table (previous exercise) with 350,000 rows.

Tasks

1. If necessary, use INSERT-SELECT to populate your Orders table from the DS.Orders table.

```
INSERT INTO Orders SELECT * FROM DS.Orders;
```

Using the DBC.TableSizeV view, what is the amount of Perm space used. Orders = _____

2. Create a new table, named "Orders_MVC", based on the Orders table except compress the following clerk names:

April Rains, Auggie Sunshine, Dee Winter, Fred Cold, Jack Snow, June Summer,
Justin Heat, Matt Winds, May Flowers, Norbert Thanks, Ossie Leaves, Steve Fall

Using INSERT-SELECT, populate your Orders_MVC table from DS.Orders table.

Using the DBC.TableSizeV view, what is the amount of Perm space used. Orders_MVC = _____

Lab Exercise 18-1 (cont.)

To turn BLC (Block Level Compression) off, you can use the following SQL:

```
SET QUERY_BAND = 'BLOCKCOMPRESSION=NO;' UPDATE FOR SESSION;
```

The following SQL can be used to determine the size of a table.

SUM of Perm space using the DBC.TableSizeV view.

```
SELECT      TableName (CHAR(15)), SUM(CurrentPerm)
FROM        DBC.TableSizeV
WHERE       DatabaseName = DATABASE
AND         TableName = 'tablename'
GROUP BY    1
ORDER BY    1;
```

To turn BLC (Block Level Compression) back on, you can use the following SQL:

```
SET QUERY_BAND = 'BLOCKCOMPRESSION=YES;' UPDATE FOR SESSION;
```

Lab Exercise 18-1 (cont.)

3. Within your session, use QUERYBAND to turn **off** block level compression.

```
SET QUERY_BAND = 'BLOCKCOMPRESSION=NO;' UPDATE FOR SESSION;
```

4. Delete the rows in your Orders and Orders_MVC tables.

5. Populate your Orders and Orders_MVC tables using the DS.Orders table.

6. Using DBC.TableSizeV, what is the amount of Perm space used. Orders = _____

7. Using DBC.TableSizeV, what is the amount of Perm space used. Orders_MVC = _____

8. Within your session, use QUERYBAND to turn **on** block level compression.

```
SET QUERY_BAND = 'BLOCKCOMPRESSION=YES;' UPDATE FOR SESSION;
```

9. Delete the rows in your Orders and Orders_MVC tables.

10. Populate your Orders and Orders_MVC tables using the DS.Orders table.

Lab Exercise 18-2

This lab exercise provides the opportunity to use the Teradata 16.10 MAPS feature.

The Nation_C table was created as follows:

```
CREATE SET TABLE PD.Nation_C, Fallback,  
  MAP = TD_MAP1  
(  
  NationKey INTEGER,  
  NationName VARCHAR(25) CHARACTER SET LATIN NOT CASESPECIFIC,  
  RegionKey INTEGER)  
UNIQUE PRIMARY INDEX (NationKey);
```

The Region_C table was created as follows:

```
CREATE SET TABLE PD.Region_C, Fallback,  
  MAP = TD_MAP1  
(  
  RegionKey INTEGER,  
  RegionName VARCHAR(25) CHARACTER SET LATIN NOT CASESPECIFIC)  
UNIQUE PRIMARY INDEX (RegionKey);
```

To specify a sparse MAP and optionally the COLOCATE option, the following SQL can be used.

```
CREATE TABLE tablename, Fallback,  
  MAP = TD_1AmpSparseMap_1Node COLOCATE USING Co_NationRegion  
( column1      INTEGER,  
   column2      VARCHAR(25),  
   column3      INTEGER)  
UNIQUE PRIMARY INDEX (column1);
```

Lab Exercise 18-2

Lab Exercise 18-2

Purpose

In this lab, you will populate tables using the sparse MAPS capability.

What you need

Populated PD.Nation_C and PD.Region_C tables.

Tasks

1. Create two Contiguous tables Nation_C and Region_C from PD.Nation_C and PD.Region_C. (Note: these tables do not have all of the nations/regions, just a small subset of data)

```
CREATE TABLE Region_C AS PD.Region_C WITH DATA;  
CREATE TABLE Nation_C AS PD.Nation_C WITH DATA;
```

2. Find the system single AMP (1) Sparse MAP by using the view DBC.MapsV.

```
SELECT MapName FROM DBC.MapsV;
```

Name of Sparse Map for 1 AMP – _____

3. Use SHOW TABLE and create new Nation_S and Region_S sparse tables specifying the 1 AMP sparse MAP name and use a COLOCATE value of Co_NationRegion for both tables (i.e., COLOCATE USING Co_NationRegion).

Lab Exercise 18-2 (cont.)

Use DBC.TableSizeV to determine which AMPs and the number of bytes that a table is currently using.

SUM of Perm space at the AMP level using the DBC.TableSizeV view.

```
SELECT DatabaseName, TableName, Vproc, SUM(CurrentPerm)
FROM DBC.TablesizeV
WHERE DatabaseName = DATABASE
AND TableName IN ('Nation_S', 'Region_S')
GROUP BY 1,2,3
ORDER BY 1,2,3;
```

SUM of Perm space for all AMPs using the DBC.TableSizeV view.

```
SELECT DatabaseName, TableName, SUM(CurrentPerm)
FROM DBC.TablesizeV
WHERE DatabaseName = DATABASE
AND TableName IN ( 'Nation_S', 'Region_S',
                  'Nation_C', 'Region_C' )
GROUP BY 1,2
ORDER BY 1,2;
```

To view the AMPs associated with your tables assigned to a sparse map, use the following view.

```
SELECT * FROM DBC.SparseMapAmpsV where DatabaseName = DATABASE;
```

To view objects that are associated with sparse maps and COLOCATE value, use the following view.

```
SELECT * FROM DBC.ObjectsInSparseMapsV;
```

Lab Exercise 18-2 (cont.)

4. Populate the two Sparse tables with INSERT-SELECT from the two contiguous tables.
5. Use the DBC.TableSizeV and determine which 2 AMPs the sparse tables are using.
Why are 2 AMPs used per sparse table? _____
6. Compare the sizes of Nation_S and Region_S to the Nation_C and Region_C tables.
Size of Nation_S = _____ Size of Nation_C = _____
Size of Region_S = _____ Size of Region_C = _____
7. Use the DBC.SparseMapAmpsV to determine which AMP has the primary data of your Sparse tables. AMP # = _____

```
SELECT * FROM DBC.SparseMapAmpsV where DatabaseName = DATABASE;
```

8. Execute the following EXPLAIN of a join of the Contiguous tables.
Is the data redistributed for the join? Yes/No _____

```
EXPLAIN  SELECT      NationName, RegionName
          FROM        Nation_C
          INNER JOIN   Region_C
          ON           Nation_C.RegionKey = Region_C.RegionKey ;
```

Execute the same EXPLAIN using the Nation_S and Region_S sparse tables. Is the data redistributed for the join? Yes/No _____

9. Optional: Create a new Nation_S2 table and a new Region_S2 table **without** using the COLOCATE option. Use the DBC.SparseMapAmpsV to determine which AMP has your Sparse tables. Are the two tables on the same AMP? Yes/No _____

Lab Exercise 20-1

Check your understanding of the concepts discussed in this module by completing the lab exercises as directed by your instructor.

Your Orders_NoPI table should have 450,000 rows from a previous exercise. If not, you can populate this table using the DS.Orders and DS.Orders_2019 tables.

SQL hint:

```
COLLECT STATISTICS COLUMN column_name [, COLUMN column_name]  
ON table_name;
```

Lab Exercise 20-1

Lab Exercise 20-1

Purpose

In this lab, you will use the COLLECT STATISTICS command.

What you need

Orders_NoPI table and populated DS.Orders and DS.Orders_2019 tables.

Tasks

1. Orders_NoPI should have 450,000 rows from a previous lab exercise. Verify the row count.
2. Collect statistics individually on the columns custid and orderdate of the Orders_NoPI table. Use the HELP STATISTICS and the HELP CURRENT STATISTICS commands on this table.

What is the row count (*) for this table? _____

How many unique values are there for custid? _____

How many unique values are there for orderdate? _____

Are the values different for HELP STATISTICS and HELP CURRENT STATISTICS? _____

3. Insert the rows from the DS.Orders table into your Orders_NoPI table. Use the HELP CURRENT STATISTICS command to answer the following questions.

What is the row count (*) for this table? _____

How many unique values are there for custid? _____

How many unique values are there for orderdate? _____

Why has the row count and number of orderdates increased with current statistics?

Lab Exercise 20-2

Check your understanding of the concepts discussed in this module by completing the lab exercises as directed by your instructor.

SQL hint:

```
COLLECT STATISTICS COLUMN column_name [, COLUMN column_name]  
ON table_name;
```

If your Orders and Orders_RP tables have a different number of rows, this is probably because Orders only has 9 years of data and Orders_RP has 10 years of data. To ensure the two tables have a similar number of rows, you can do either of the following.

One option is to copy the rows from Orders_2019 into the Orders table.

```
INSERT INTO Orders SELECT * FROM Orders_2019;
```

Another option is to delete all the rows in Orders table and populate the Orders table from Orders_RP.

```
DELETE Orders;  
INSERT INTO Orders SELECT * FROM Orders_RP;
```

Lab Exercise 20-2

Lab Exercise 20-2

Purpose

In this lab, you will EXPLAIN various SQL access functions.

What you need

Populated tables from a previous lab. Execute the following INSERT-SELECT so both the Orders and Orders_RP tables are similar in size.

```
INSERT INTO Orders SELECT * FROM Orders_2019;
```

Tasks

1. Collect statistics on orderid and orderdate for the Orders and Orders_RP tables. Collect statistics on PARTITION for the Orders_RP table. Execute the following two Explains.

```
EXPLAIN SELECT * FROM Orders  
WHERE orderdate BETWEEN DATE '2019-01-01' AND DATE '2019-01-31';
```

How many AMPs are accessed in this operation? _____

Is this a full table scan? _____ What type of lock is used? _____

```
EXPLAIN SELECT * FROM Orders_RP  
WHERE orderdate BETWEEN DATE '2019-01-01' AND DATE '2019-01-31';
```

How many AMPs are accessed in this operation? _____

Is this a full table scan? _____ What type of lock is used? _____

Is partitioning beneficial for this type of query? _____

Lab Exercise 20-2 (cont.)

Check your understanding of the concepts discussed in this module by completing the lab exercises as directed by your instructor.

Lab Exercise 20-2 (cont.)

2. Execute the following three Explains.

```
EXPLAIN SELECT * FROM Orders
      WHERE orderid = 1404662;
```

How many AMPs are accessed in this operation? _____

How is this row retrieved by Teradata? _____

```
EXPLAIN SELECT * FROM Orders_RP
      WHERE orderid = 1404662;
```

How many AMPs are accessed in this operation? _____

How many partitions are scanned to locate this row? _____

How is this row retrieved by Teradata? _____

```
EXPLAIN SELECT * FROM Orders_NoPI
      WHERE orderid = 1404662;
```

How many AMPs are accessed in this operation? _____

How is this row retrieved by Teradata? _____

Lab Exercise 20-2 (cont.)

Check your understanding of the concepts discussed in this module by completing the lab exercises as directed by your instructor.

Lab Exercise 20-2 (cont.)

3. Execute the following two Explains.

```
EXPLAIN DELETE FROM Orders
      WHERE orderdate BETWEEN DATE '2019-01-01' AND DATE '2019-12-31';
```

How many AMPs are accessed in this operation? _____

Is this a full table scan? _____ Why or why not? _____

```
EXPLAIN DELETE FROM Orders_RP
      WHERE orderdate BETWEEN DATE '2019-01-01' AND DATE '2019-12-31';
```

How many AMPs are accessed in this operation? _____

Is this a full table scan? _____ Why or why not? _____

How many partitions are scanned to delete these rows? _____

4. Collect statistics on orderid for the Orders_CP table and execute the following two Explains.

```
EXPLAIN SELECT orderid, custid FROM Orders_CP;
```

How many column partitions are accessed? _____

What is the relative cost for this EXPLAIN? _____

```
EXPLAIN SELECT orderid, custid, orderstatus, orderdate FROM Orders_CP;
```

How many column partitions are accessed? _____

What is the relative cost for this EXPLAIN? _____

Why is there an extra partition accessed for each of these explains? _____

Lab Exercise 20-2 (cont.)

Check your understanding of the concepts discussed in this module by completing the lab exercises as directed by your instructor.

Lab Exercise 20-2 (cont.)

4. (cont.) Last of the three Explain plans.

```
EXPLAIN SELECT * FROM Orders_CPPI WHERE custid = 1024;
```

How many AMPs are involved in this query? _____

How many partitions are involved in this query? _____

Is this number equal to the number of columns in the table? _____

Why or why not? _____

Lab Exercise 23-1

Check your understanding of the concepts discussed in this module by completing the lab exercises as directed by your instructor.

Use the following SQL for this exercise.

To populate a table:

```
INSERT INTO tablename1  
SELECT * FROM tablename2;
```

The Explain from the facing page is listed below for your convenience and can be pasted into your query window.

```
EXPLAIN  
SELECT      LastName, FirstName, DeptName, JobDesc  
FROM        Employee E  
INNER JOIN  Department D ON E.DeptNumber = D.DeptNumber  
INNER JOIN  Job J      ON E.JobCode = J.JobCode  
ORDER BY    3, 1, 2;
```

Lab Exercise 23-1

Lab Exercise 23-1

Purpose

In this lab, you will create join indexes and evaluate the Explains of various joins.

What you need

Populated PD tables and Employee, Department, and Job tables in your database

Tasks

1. Verify the number of rows in Employee, Job, and Department tables. If not correct, use INSERT/SELECT to populate the tables from the PD database.

Employee	Count = 150,000
Department	Count = 17,415
Job	Count = 11,124

2. Execute the following EXPLAIN statement.

```
EXPLAIN
SELECT      LastName, FirstName, DeptName, JobDesc
FROM        Employee E
INNER JOIN  Department D      ON  E.DeptNumber = D.DeptNumber
INNER JOIN  Job J            ON  E.JobCode = J.JobCode
ORDER BY    3, 1, 2;
```

What is the total estimated time cost for this EXPLAIN? _____

Lab Exercise 23-1 (cont.)

Check your understanding of the concepts discussed in this module by completing the lab exercises as directed by your instructor.

Use the following SQL for this exercise.

To create a join index:

```
CREATE JOIN INDEX join_index_name AS  
SELECT ...;
```

SUM of Perm space using the DBC.TableSizeV view.

```
SELECT      TableName (CHAR(15)), SUM(CurrentPerm)  
FROM        DBC.TableSizeV  
WHERE       DatabaseName = DATABASE  
AND         TableName = 'join_index_name'  
GROUP BY    1  
ORDER BY    1;
```

Lab Exercise 23-1 (cont.)

3. Create a “non-compressed” join index which includes the following columns of Employee, Department, and Job.

Employee	- LastName, FirstName
Department	- DeptNumber, DeptName
Job	- JobCode, JobDesc

Execute the HELP USER command. What is the object type of the Join Index? _____

4. Execute the following EXPLAIN (same SQL statement as step #2).

```

EXPLAIN
SELECT      LastName, FirstName, DeptName, JobDesc
FROM        Employee E
INNER JOIN  Department D ON  E.DeptNumber = D.DeptNumber
INNER JOIN  Job J          ON  E.JobCode = J.JobCode
ORDER BY   3, 1, 2;
```

What is the total estimated time cost for this EXPLAIN? _____

Is the join index used? _____

How much space does the join index require (use the DBC.TableSizeV view)? _____

Lab Exercise 23-1 (cont.)

Check your understanding of the concepts discussed in this module by completing the lab exercises as directed by your instructor.

The Explain from the facing page is listed below for your convenience and can be pasted into your query window.

```
EXPLAIN  
SELECT          LastName, FirstName, DeptName, JobDesc,  
                  SalaryAmount  
FROM            Employee E  
INNER JOIN      Department D ON  E.DeptNumber = D.DeptNumber  
INNER JOIN      Job J          ON  E.JobCode = J.JobCode  
ORDER BY        3, 1, 2;
```

Lab Exercise 23-1 (cont.)

5. Execute the following EXPLAIN statement – SalaryAmount has been added as a projected column.

```
EXPLAIN
SELECT      LastName, FirstName, DeptName, JobDesc, SalaryAmount
FROM        Employee E
INNER JOIN   Department D ON  E.DeptNumber = D.DeptNumber
INNER JOIN   Job J          ON  E.JobCode = J.JobCode
ORDER BY    3, 1, 2;
```

What is the total estimated time cost for this EXPLAIN? _____

Is the join index used? _____ If not, why not? _____

6. Drop the Join Index.
7. Create a second “non-compressed” join index similar to step #3 except include the Primary Index of EmpNumber for the Join Index.

```
Employee      – LastName, FirstName, EmpNumber
Department    – DeptNumber, DeptName
Job           – JobCode, JobDesc
```

8. Execute the EXPLAIN statement from step #5.

What is the total estimated time cost for this EXPLAIN? _____

Is the join index used? _____ If so, why? _____

9. Drop the Join Index.

Lab Exercise 23-2 (optional)

Check your understanding of the concepts discussed in this module and the columnar module by completing the lab exercises as directed by your instructor.

Use the following SQL for this exercise.

To populate a table:

```
INSERT INTO tablename1  
SELECT * FROM tablename2;
```

Lab Exercise 23-2 (optional)

Lab Exercise 23-2 (optional)

Purpose

In this lab, you will create join indexes using Columnar and evaluate the various affects.

What you need

Populated PD tables and Employee, Department, Job and EmpPhone tables in your database

Tasks

1. Verify the number of rows in Employee, Job, Department and EmpPhone tables. If not correct, use INSERT/SELECT to populate the tables from the PD database.

Employee	Count = 150,000	Job	Count = 11,124
Department	Count = 17,415	EmpPhone	Count = 350,000

2. Execute the following Explain statement which joins all 4 tables together.

```
EXPLAIN
SELECT      D.DeptName, E.EmpNumber, E.LastName, E.FirstName,
            J.JobDesc, P.AreaCode, P.PhoneNumber
FROM        Employee E
INNER JOIN   Department D ON      E.DeptNumber = D.DeptNumber
INNER JOIN   EmpPhone P  ON P.EmpNumber = E.EmpNumber
INNER JOIN   Job J       ON E.JobCode = J.JobCode
ORDER BY    1, 2, 3, 4;
```

What is the total estimated time cost for this EXPLAIN? _____

Lab Exercise 23-2 (cont.)

Check your understanding of the concepts discussed in this module by completing the lab exercises as directed by your instructor.

Use the following SQL for this exercise.

To create a join index:

```
CREATE JOIN INDEX  employee_JI AS
SELECT  EmpNumber,
        DeptNumber,
        EmpMgrNumber,
        JobCode,
        LastName,
        FirstName,
        SalaryAmount,
        ROWID AS RID
FROM    Employee
PRIMARY INDEX (DeptNumber);
```

SUM of Perm space using the DBC.TableSizeV view.

```
SELECT      TableName (CHAR(15)), SUM(CurrentPerm)
FROM        DBC.TableSizeV
WHERE       DatabaseName = DATABASE
AND         TableName = 'join_index_name'
GROUP BY    1
ORDER BY    1;
```

The Explain from the facing page is listed below for your convenience and can be pasted into your query window.

```
EXPLAIN
SELECT      D.DeptName, E.EmpNumber, E.LastName, E.FirstName,
            J.JobDesc, P.AreaCode, P.PhoneNumber
FROM        Employee E
INNER JOIN  Department D ON E.DeptNumber = D.DeptNumber
INNER JOIN  EmpPhone P   ON P.EmpNumber = E.EmpNumber
INNER JOIN  Job J        ON E.JobCode = J.JobCode
ORDER BY    1, 2, 3, 4;
```

Lab Exercise 23-2 (cont.)

3. Create the following Global Join Index on the employee table.

```
CREATE JOIN INDEX Employee_JI AS
SELECT      EmpNumber,
            DeptNumber,
            EmpMgrNumber,
            JobCode,
            LastName,
            FirstName,
            SalaryAmount,
            ROWID AS RID
FROM        Employee
PRIMARY INDEX (DeptNumber);
```

4. Re-execute the Explain statement from step #2

What is total estimated time cost for this EXPLAIN? _____

Is the join index used? _____

How much space does the join index require (use the DBC.TableSizeV view)? _____

5. Drop the Employee_JI Join Index.

Lab Exercise 23-2 (cont.)

Check your understanding of the concepts discussed in this module by completing the lab exercises as directed by your instructor.

Hint: To create a column partitioned join index, you can leverage the following syntax:

To create a column partitioned join index with a PRIMARY AMP option:

```
CREATE JOIN INDEX employee_JI_CPPA AS
SELECT  EmpNumber,
        DeptNumber,
        EmpMgrNumber,
        JobCode,
        LastName,
        FirstName,
        SalaryAmount,
        ROWID AS RID
FROM    Employee
PRIMARY AMP INDEX (DeptNumber) PARTITION BY COLUMN;

/* PRIMARY AMP INDEX or PRIMARY AMP can be used */
```

To create a column partitioned join index with a PRIMARY INDEX option:

```
CREATE JOIN INDEX employee_JI_CPPI AS
SELECT  EmpNumber,
        DeptNumber,
        EmpMgrNumber,
        JobCode,
        LastName,
        FirstName,
        SalaryAmount,
        ROWID AS RID
FROM    Employee
PRIMARY INDEX (DeptNumber) PARTITION BY COLUMN;
```

Lab Exercise 23-2 (cont.)

6. Modify the Create Join Index you used in step #3 and make a Column Partitioned Join index called **Employee_JI_CPPA** with a Primary AMP Index of **DeptNumber** column.
7. Re-execute the Explain statement from step #2.
 What is total estimated time cost for this EXPLAIN? _____
 What is the size Employee_JI_CPPA join index is using? _____
8. Drop the Employee_JI_CPPA Join Index.
9. Modify the Create Join Index you used in step #6 and make a Column Partitioned Join index called **Employee_JI_CPPI** with a Primary Index of **DeptNumber** column.
10. Re-execute the Explain statement from step #2.
 What is total estimated time cost for this EXPLAIN? _____
 What is the size Employee_JI_CPPI join index is using? _____
11. Drop the Employee_JI_CPPI Join Index.
12. Which Join Index gave the best tradeoff with regards to size and performance? _____

Notes